MIL-W-52572D
7 August 1978
SUPERSEDING
MIL-W-52572C
29 January 1973

# MILITARY SPECIFICATION

# WATER PURIFICATION UNIT, ELECTRIC-MOTOR-DRIVEN, ERDLATOR-TYPE CLARIFIER, DIATOMITE-TYPE FILTER, 420 GALLONS PER HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a military design electric-motor-driven, 420-gph, water purification unit.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-V-173 | - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment). |
| MIL-P-514 | - Plates, Identification, Instruction and Marking Blank. |

FSC 4610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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| | |
|---|---|
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |
| MIL-S-52333 | - Sleeves, Filter Element; Water Purification Units. |
| MIL-W-52571 | - Water Purification Unit, and Water Purification Equipment Set, 420-GPH, Packaging of. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |
| MIL-STD-1472 | - Human Engineering Design Criteria for Military Systems, Equipment and Facilities. |
| MIL-STD-1474 | - Noise Limits for Army Materiel. |
| MS16203 | - Nut, Plain, Hexagon, and Hexagon Jam, UNC-2B, Nonmagnetic. |
| MS35307 | - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Corrosion Resisting, Passivated, UNC-2A. |
| MS35842 | - Clamp, Hose: Low Pressure Type "F". |

DRAWINGS

ME

| | |
|---|---|
| TA13215E7750 | - Water Purification Unit, 420 GPH, Assembly. |
| D13208E5820 | - Terminal, Load. |

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.
A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.9 - Corrosion-Resisting Chromium and Chromium-Nickel Steel Welding Rods and Bare Electrodes.
A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX - Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A123 - Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars and Strip.
A575 - Merchant Quality Hot-Rolled Carbon Steel Bars.
A576 - Special Quality Hot-Rolled Carbon Steel Bars.
B173 - Rope-Lay Stranded Copper Conductors Having Concentric-Stranded Members for Electrical Conductors.
B209 - Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The water purification unit shall be in accordance with Top Assembly TA13215E7750 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more water purification units, as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce water purification units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2) the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more water purification units for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings (see 6.13). Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The material selected shall have no adverse effect on health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting officer to the appropriate medical service who will act as an advisor to the contracting officer (see 6.4).



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3.5 Accessories. The accessories specified in Table I shall be furnished in the quantities shown.

Table I. Accessories

| Item | Quantity for each unit |
|---|---|
| Flange, blind; conforming to Figure 1 | 4 |
| Cap, screw; hexagon head (finished hexagon bolt), MS35307-359. | 8 |
| Clamp, hose; low-pressure, filter element MS35842-15. | 8 |
| Ground rod; conforming to Figure 2. | 2 |

3.6 Lubrication. All moving parts, not in contact with water being treated, requiring lubrication shall be provided with a means for lubricating.

3.6.1 Lubricants. The water purification unit shall operate as specified herein when lubricated with military lubricants (see 6.5).

3.6.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.6.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.6.4 Filling, draining, and checking provisions. Enclosures such as gear cases that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain shall be fitted with a drain plug conforming to SAE J531. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Integral tubes or troughs shall be used to convey the lubricant from the drain to a point where it can be collected in a container. The drain plug, the filling means, and the lubricant level dipsticks or holes shall be accessible without disturbance of accessories or parts.



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3.6.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, electronic gear, and mechanical control systems; for use on gears, actuator screws, and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

(b) MIL-G-10924 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of minus 65° F to plus 225° F. Not for use on machinery which comes in contact with potable water.

The water purification unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.6.6 Oil lubrication. Unless otherwise preserved for packaging, all components, with the exception of the chemical solution feeder, shall be filled to the operating level with grades of military oils designated for use in temperatures above 32° F (see 6.5). Each unit shall be tagged in a conspicuous place to indicate the grades of lubricants used.

3.7 Electromagnetic interference. The electromagnetic interference emission characteristics of the water purification unit shall conform to MIL-STD-461, for electrical/electromechanical equipment category (see 6.6).

3.8 Fungus and moisture resistance. The electric circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with varnish conforming to MIL-V-173, composition as specified in 3.8.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant, or which are hermetically sealed, need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.



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When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.8.1 Composition. Composition I shall be used unless Composition II is required in order to comply with local air pollution regulations governing the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.9 Identification marking. The water purification unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the water purification unit on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. The plates shall be attached to the water purification unit with adhesive and located as specified on the applicable drawing.

3.10 Shipping data plate. A shipping data plate conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material shall be furnished. The plate shall show the silhouette of the water purification unit in transport position, indicating the location of the lifting and the tiedown attachments. The plate shall be attached to the water purification unit with rivets, screws, or adhesive and located as specified on the applicable drawing.

3.11 Reliability. The specified Mean-Time-Between-Failure (MTBF) of the water purification unit and ancillary equipment shall be 198 hours when tested as specified in 4.6.2.6.

3.12 Maintainability. Each maintenance assembling and disassembling operation performed as a result of testing in accordance with 4.6.2.7 shall be accomplished by not more than two men using common tools furnished with the water purification unit. The ratio of manhours of maintenance required to the hours of operation shall not exceed 0.08. A maintenance schedule shall be furnished prior to start of any testing.

3.13 Government-furnished property. Unless otherwise specified, (see 6.2), four filter element sleeves conforming to MIL-S-52333, Size 3, will be furnished by the Government for each water purification unit (see 6.7).



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3.14 Noise level. The noise level of the water purification unit (excluding horns and warning alarms) shall be in accordance with MIL-STD-1474, Category D, for steady-state noise when tested in accordance with 4.6.2.8.

3.15 Workmanship. All parts, components, and assemblies of the water purification units including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other extraneous material. Unless otherwise specified in drawings, all edges shall be rounded or chamfered. All parts, components, and assemblies of the water purification units shall be free from misalinement, incomplete welds, leaks, and other defects that could impair the operation and reliability of the unit.

3.15.1 Castings and forgings. All parts, components, and assemblies of the water purification unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.15.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.15.3 Welders, weldments, and welding.

3.15.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.



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Welders who make only horizontal welds need not qualify for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.15.3.2 Weld filler metals. Aluminum filler metal shall be in accordance with AWS A5.10, Type ER5356 or ER5556 when joining base metal Type 6061 to other types of aluminum. Steel filler metal shall be in accordance with AWS A5.9, Type ER308L for corrosion-resisting steel and AWS A5.1, Type E6010 or E6011 for carbon steel.

3.15.3.3 Aluminum and steel weldment. The surfaces and edges of parts to be welded including layers of weld materials when multiple layers are required, shall be free from oxide scale, paint, lubricant and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities and other imperfections that could reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified on the applicable drawing.

3.15.3.3.1 Aluminum welding. Aluminum welding, except spot and seam welding, shall be done by the metal-inert-gas (MIG) method or pulse arc method that will provide equivalent mechanical properties of the filler metal to base metal combinations as specified in 3.15.3.2. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic and nonmetallic inclusions. Unless otherwise specified on the drawings, finished aluminum welds shall not be ground, filed, chipped, or hammered, but may be wire-brushed. Preheating for welding is permissible provided the temperature does not exceed 300° F for a total time not to exceed 30 minutes. No process requiring the use of welding flux shall be used. Work shall be positioned and jigged for flat welding whenever practicable.

3.15.3.3.2 Steel welding. Steel welding shall be done by shielded-electric-arc method or metal-inert-gas (MIG) process in accordance with qualification procedures specified in 3.15.3.1, or by any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations as specified (see 3.15.3.2). Where embrittlement of the joined metal may result, preheating or annealing or both shall be



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employed. Work shall be positioned for flat welding whenever practicable. Complete and uniform penetration and fusion of the metals shall be obtained on all welds.

3.15.3.4 Weldment specimens. The contractor shall submit steel and aluminum specimens representing each typical weld joint utilized in fabrication of the water purification unit. These specimens shall be furnished at the same time as the preproduction water purification unit. Weldment specimens shall be submitted by each welder performing weldments on the water purification unit with each type of welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as an inspection standard for evaluation of production weldments.

3.15.3.5 Welding procedure. Prior to the fabrication of any weldment, the contractor shall prepare and submit to the contracting officer for approval a written welding procedure defining process and control. This procedure shall include, but is not limited to, those factors listed in Table II. Welding procedure changes affecting any of the factors in Table II shall be subject to approval of the contracting officer prior to adoption of the change.

Table II. Factors to be Included in Joint Welding Procedures

| | |
|---|---|
| 1. | Welding process and techniques including shielding medium. |
| 2. | Welding process operation. |
| 3. | Type of current, ac or dc, and polarity of dc if used. |
| 4. | Shielding gas composition. |
| 5. | Joint design. |
| 6. | Method of joint edge preparation. |
| 7. | Preweld and interpass cleaning. |
| 8. | Preheat or ambient temperature range. |
| 9. | Composition of filler metal. |
| 10. | Filler metal size for all passes. |
| 11. | Number and sequence of passes. |
| 12. | Welding position. |
| 13. | Repair method to be used in removing defects. |
| 14. | Method of inspection used to insure correct repair. |
| 15. | Method of qualification of welders. |
| 16. | Method for determining capability of automatic equipment and equipment operators. |
| 17. | Workmanship specimens. |
| 18. | Method of inspection of finished welds. |



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3.15.3.6 Rewelding defects. Portions of welded joints that have been rejected on inspection because of defects or nonconformance to weldments specimens of 3.15.3.4 may be repaired only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Repairs in the form of patching aluminum shall not be performed.

3.15.3.7 Jigs and fixtures. Components except minor parts shall be assembled in jigs or frames and joined while held in position. Jigs or frames shall be designed to minimize distortion of the parts being welded.

3.15.4 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall be applied to give a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

3.15.5 Cleaning. The water purification unit and components shall be thoroughly cleaned of all cutting oil, residue of tests, grease, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the water purification unit and components unsuitable for its intended use. The water purification unit shall be completely drained before packaging.

3.16 Transportability. The water purification unit shall be capable of withstanding the shock and vibrational stress encountered during transportation.

3.17 Environmental.

3.17.1 Operating temperature. The water purification unit shall perform as specified in ambient temperatures from +110° F to +32° F.

3.17.2 Storage temperature. The water purification unit shall not be damaged by storage at ambient temperatures from +160° F to -30° F.

3.18 Safety. A grounding system shall be incorporated to insure safety from static electricity.

3.19 Human factors. The characteristics of the water purification unit shall provide for operation by personnel in all type clothing, and shall conform to MIL-STD-1472.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Water purification units which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards, and applicable drawings.

4.1.3 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.4 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan and the basis for acceptance and rejection (see 6.8 and 6.9.1).

4.1.5 End Item Final Inspection Requirement (EIFIR). The preproduction model and each production unit shall be inspected in accordance with EIFIR-EF13215E7750. One copy of the completed EIFIR shall be shipped with each inspected water purification unit (see 6.9.2).



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction water purification unit shall be examined for defects marked "X" in Column 1 of Table III. Presence of one or more defects shall be cause for rejection or shall be cause for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests. The preproduction water purification unit shall be subjected to tests marked "X" in Column 1 of Table IV. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.3.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production water purification units will be selected at random by the Government from water purification units being produced from production tooling and will be examined for the defects marked "X" in Column 2 of Table III and subjected to tests marked "X" in Column 2 of Table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the initial production water purification unit shall not exclude the remaining water purification units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial producation test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production test.

4.4.1 Inspection failure. Failure of an initial production water purification unit to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production water purification unit(s) and shall be cause for refusal by the Government to continue acceptance of production water purification units until evidence has been provided by the contractor that corrective action has been taken to eliminate the



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deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all water purification units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each water purification unit shall be examined for the defects marked "X" in Column 3 of Table III. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Upon successful completion of the examination 4.5.1, each water purification unit shall be subjected to the tests marked "X" in Column 3 of Table IV. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table III.

Table III. Examination Schedule

| Pre-Prod. | Init. Prod. | Qual. Conf. | Defect | Requirement Paragraph |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| X | X | - | 101. Nonconformance to dimensions, assembly, and construction details as shown on applicable drawing, that can be checked without disassembling of any component. | 3.1, 3.1.1 |
| X | - | - | 102. Material not as specified. | 3.4 |
| X | X | - | 103. Accessories not as specified. | 3.5 |
| X | X | - | 104. Lubricants not as specified. | 3.6.1 |
| X | X | - | 105. Location and type of lubrication fittings not as specified. | 3.6.2 |
| X | X | - | 106. Pressure-release device not as specified. | 3.6.3 |
| X | X | - | 107. Filling, draining, and checking provisions not as specified. | 3.6.4 |



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Table III. Examination Schedule (Cont'd)

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| X | X | - | 108. Grease lubrication not as specified. | 3.6.5 |
| X | - | - | 109. Oil lubrication not as specified. | 3.6.6 |
| X | - | - | 110. Nonconformance to fungus and moisture resistance. | 3.8 |
| X | X | X | 111. Identification marking illegible or not attached as specified. | 3.9 |
| X | X | X | 112. Shipping data plates missing, attached or illegible, as specified. | 3.10 |
| X | - | X | 113. Workmanship not as specified. | 3.14 |

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to test, the water purification unit shall be lubricated with greases and oils specified in 3.6. Oils shall be those designated for use in the ambient temperature at the place of test.

4.6.2.2 Test schedule. Tests shall be conducted in accordance with Table IV.

Table IV. Test Schedule

| Pre-Prod. | Init. Prod. | Qual. Conf. | Tests | Test Paragraph | Requirement |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| X | - | - | Electromagnetic interference. | 4.6.2.3 | 3.7 |
| X | X | - | Operational (6 hr.). | 4.6.2.4 | 3.2, 3.2.2, 3.3 |
| - | - | X | Operational (2 hr.). | 4.6.2.5 | 3.2, 3.2.2, 3.3 |
| - | X | - | Reliability (MTBF). | 4.6.2.6 | 3.11 |
| - | X | - | Maintainability. | 4.6.2.6 | 3.11 |
| - | X | - | Noise level. | 4.6.2.8 | 3.14 |
| - | X | - | Transportability. | 4.6.2.9 | 3.16 |
| - | X | - | Environmental. | 4.6.2.10 | 3.17 |
| - | X | - | Safety. | 4.6.2.11 | 3.18 |
| - | X | - | Human factors. | 4.6.2.12 | 3.19 |



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4.6.2.3 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.7. The contractor shall furnish the contracting officer a report of test as required by MIL-STD-461 (see 6.6). Disapproval of the report shall constitute failure of this test.

4.6.2.4 Operational (6 hours). The preproduction model water purification unit shall be operated with tap water for not less than 6 hours as specified in 4.6.2.4.1 through 4.6.2.4.6.

4.6.2.4.1 Test equipment. Test equipment shall include but shall not be limited to the following:

Power source - 115-volt, 1-phase, 60-Hertz.
Cable, electrical - lengths as specified in the applicable drawing.
Hose, discharge - two lengths of 1-inch I.D. hose with 11-1/2 NPSH connectors.
Water - Tap (potable).

4.6.2.4.2 Preparation of water purification unit. Prepare the water purification unit for the operational test as follows:

(a) Assemble all components of the water purification unit on the same plane.
(b) Connect the water purification unit to the power source.
(c) Connect one length of discharge hose from the raw-water pump effluent to the raw-water influent of the water purification unit.
(d) Connect all other applicable equipment drain hoses to their respective outlets of the water purification unit, and hook effluent hoses in their respective places.
(e) Connect each component to its respective electrical outlet of the electrical control box.
(f) Connect the concentrator tank to the erdlator tank with the applicable equipment hose.
(g) Connect one length of applicable equipment hose from the effluent of the collapsible fabric tank to the influent of the distribution pump.
(h) Connect one length of discharge hose from the effluent of the distribution pump, and attach dispensing nozzle.
(i) Fill both chemical solution containers with tap water and insert chemical solution feeders influent tubing into their respective containers.



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4.6.2.4.3 Procedure for starting water-treating section. The procedure for starting the water-treating section shall be as follows:

(a) Start air compressor.
(b) Start raw-water pump to fill erdlator tank with tap water. The chemical slurry tanks may be filled simultaneously.
(c) As soon as raw water enters the erdlator, start the chemical solution feeders.
(d) When erdlator tank is half full, start erdlator agitator motor.
(e) While erdlator tank is filling, prepare the filter section for operation as specified in 4.6.2.4.4.
(f) Turn on bell alarm switch for chemical solution feeders.

4.6.2.4.4 Preparation of the filter section. The preparation of the filter section shall be as follows:

(a) Connect one length of applicable equipment hose from appropriate wet well outlet to filter pump influent.
(b) Connect one length of applicable equipment hose from filter dome flow control valve to influent of wet well compartment.
(c) Connect one length of applicable equipment hose from effluent of the filter, to filtered water collapsible storage tank.
(d) Connect one length of applicable equipment drain hose from filter effluent drain valve to waste.
(e) Filter element sleeves are not required for tests specified herein.

4.6.2.4.5 Procedure for starting and backwashing filter section. The procedure for starting and backwashing the filter section shall be as follows:

(a) Starting.

(1) Close filter air release valve.
(2) Close filter drain valve.
(3) Close filter precoat filling pipe with pipe cap.
(4) Open filter air vent valve.
(5) Turn effluent plug valve to recirculation operation.
(6) Start filter pump.
(7) At the instant that water starts to escape from filter vent valve, close filter vent valve rapidly and open filter effluent valve.



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(8) Allow filtered water to recirculate to wet well for not less than 5 minutes; then change valve setting from recirculation to discharge to the filtered water storage tank.

(b) Backwashing.

(1) Turn filter effluent plug valve to backwash position.
(2) Allow filter pump to operate until maximum pressure is reached and water starts to escape from safety-relief valve.
(3) Stop filter pump and open filter air release valve.
(4) Open filter effluent drain valve.

Operate filter for not less than twelve 1/2-hour filtering and backwashing cycles while filling the filtered water storage tank.

4.6.2.4.6 Procedure for operating distribution pump. The procedure for operating the distribution pump shall be as follows:

(a) When filtered water storage tank is full, start distribution pump.
(b) Open dispensing nozzle.
(c) When tank has been emptied, stop pump.
(d) Operate the distribution pump for not less than four cycles.

4.6.2.4.7 Criteria for failure. Evidence of any one of the following shall constitute failure of this test:

(a) Float valve mechanism not functioning correctly.
(b) Safety-relief valve not functioning as specified.
(c) Chemical solution feeders not performing as specified.
(d) Any pump not performing as specified.
(e) Evidence of air escaping air-release valve when valve is closed.
(f) Evidence of water escaping through air-release valve during backwash cycle.
(g) Any valve not functioning as required.
(h) Air compressor not performing as specified on applicable drawing.
(i) Incorrect motor or pump rotation as specified on applicable drawing.
(j) Evidence of any leak in filtered water storage tank.
(k) Evidence of any leak from any water purification component, seam, or any connection.



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(l) Mechanical failure of any part.
(m) Slurry feeder timer not functioning correctly.
(n) Evidence of misalinement of agitator.

4.6.2.5 Operational (2 hours). Each production water purification unit shall be tested as specified in 4.6.2.4.1 through 4.6.2.4.6, except operate the complete unit for only four 1/2-hour filter starting and backwashing cycles, in lieu of the 12 cycles in the 6-hour operational test.

4.6.2.5.1 Criteria for failure. Evidence of any one of conditions specified in 4.6.2.4.7 shall constitute failure of this test.

4.6.2.6 Reliability. Using the MTBF specified in 3.11, the water purification units shall be tested as specified herein, with "accept" and "reject" criteria in accordance with Figure 3. Continue testing until either an "accept" or "reject" decision is reached. Failure, for the purpose of this test, is defined as any malfunction which the operators cannot remedy by adjustment, repair, or replacement action within 30 minutes (excluding waiting or start-up time), utilizing the resources available to the parent unit or organization when deployed, and which causes or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance capabilities of the system.
(b) Serious damage to the system by continued operation.
(c) Serious personnel safety hazard.

Simultaneous related malfunctions are considered as one failure.

4.6.2.6.1 Test procedure. The water purification unit with ancillary equipment shall be set up, operated, and maintained in accordance with the applicable technical manual supplied with the unit. The ancillary equipment shall consist of all items listed in the latest issue of SC 4610-97-CL-E13. Commence operation using the procedures outlined in 4.6.2.4.3 and 4.6.2.4.5. Refer to the technical manual for complete operational instructions in order to simulate actual use. Operate equipment continuously for periods of 20 hours per day at rated capacity, then shut down unit. Daily preventive maintenance and cleaning is allowed during the 4-hour nonoperating period (not to be included in accessing MTBF).



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4.6.2.6.2 Failure criteria. Nonconformance to 3.11 shall constitute failure of this test.

4.6.2.7 Maintainability. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.12. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.8 Noise level.

4.6.2.8.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise while the unit is operating at rated capacity. Nonconformance to 3.14 shall constitute failure of this test.

4.6.2.8.2 Exterior spectator position. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.14 shall constitute failure of this test.

4.6.2.9 Transportability. Water purification units shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure VI, and Method 514.2, Procedure XIII, Part 2, except the item shall be impacted once in each direction at a speed of 8 mph ± 5 percent (total of two impacts). Any shifting of contents, any loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the water purification unit shall constitute failure of this test.

4.6.2.10 Environmental.

4.6.2.10.1 High temperature. The water purification units shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The maximum test temperature for storage shall be 160° F and for operating shall be 125° F. Filter element sleeves shall be installed in the filter in accordance with the applicable technical manual furnished with the unit. Operation shall be in accordance with the 2-hour operational test specified in 4.6.2.5. One or more of the following shall constitute failure of this test:

(a) Leaks at any welded seam or connection.
(b) Lights, buzzer, or any switch fails to operate.
(c) Valves fail to operate properly.
(d) Evidence of water or air escaping through air-release valve or vent valve.
(e) Mechanical or electrical failure of any component or part.
(f) Cracks or splits in any filter element sleeve.



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4.6.2.10.2 Low temperature. The water purification units shall be tested in accordance with MIL-STD-810, Method 502, Procedure I. The minimum test temperatures for storage shall be -30° F, and the minimum operating temperatures shall be +32° F. Operation shall be in accordance with the 2-hour operational test specified in 4.6.2.5. One or more of the following defects shall constitute failure of this test:

(a) Leaks at any welded seam or any connection.
(b) Lights, buzzer, or any switch fails to operate.
(c) Valves fail to operate properly.
(d) Evidence of water or air escaping through air-release valve or vent valve.
(e) Mechanical or electrical failure of any component or part.
(f) Cracks or splits in any filter element sleeve.

4.6.2.11 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.18 shall constitute failure of this test.

4.6.2.12 Human factors. The water purification units shall be evaluated to determine the human factor characteristics. Nonconformance to 3.19 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select water purification units at any time during the contract production period and subject these water purification units to the examination marked "X" in Column 2 of Table III and the tests marked "X" in Column 2 of Table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on water purification units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production water purification units. In addition to any test specified as part of the inspection comparison, the tained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison water purification unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison water purification unit and shall be cause for refusal by the Government to continue acceptance of production water purification units until evidence has been provided



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by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on water purification units previously accepted and produced under the contract. Any deficiencies found a a result of the inspection comparison will be considered prima facie evidence that all water purification units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packing, and marking for Level A shall be examined to determine compliance with the Quality Assurance Provisions of MIL-W-52571. The preservation, packing, and marking for Commercial shall be examined to determine compliance with MIL-STD-1188.

## 5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level A or Commercial as specified (see 6.2). Level A preservation, packing, and marking shall be in accordance with MIL-W-52571. Commercial preservation, packing, and marking shall be in accordance with MIL-STD-1188.

## 6. NOTES

6.1 Intended use. The military design water purification unit is intended for supplying 420 gph of potable water from a raw, fresh-water source.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of preproduction models required and time frame required for submission of preproduction model (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the pre-production examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production testing is required and number of initial production water purification units to be furnished, when applicable (see 3.3).



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(e) When the Government will not furnish the equipment listed in 3.13 (see 6.7).
(f) Degree of preservation and degree of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish water purification units conforming to this specification.

6.4 Medical advisor. For Army procurement, the Surgeon General of the Army will act as advisor to make the necessary determination on health hazards (see 3.4).

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the water purification unit covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.6 Electromagnetic interference. The contracting officer should forward a copy of the test report (see 4.6.2.3) to the Commanding Officer, US Army Mobility Equipment Research and Development Command, ATTN: DRDME-GE, Fort Belvoir, VA 22060, for approval.

6.7 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.13 when applicable.

6.8 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrective action taken when applicable."

6.9 Definitions.

6.9.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected



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on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.9.2 End Item Final Inspection Requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance, and shall reflect all approved Engineering Change Proposals (ECP's), Request for Waivers (RFW's), and Request for Deviations (RFD's).

6.10 This water purification unit is a component of the Water Purification Equipment Set: 420 Gallons Per Hour; in accordance with Department of the Army Supply Catalog SC 4610-97-CL-E13.

6.11 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each water purification unit.

6.12 Provisioning. The contracting officer should include provisioning requirements for such repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of water purification units.

6.13 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodians:
Army - ME
Air Force - 99

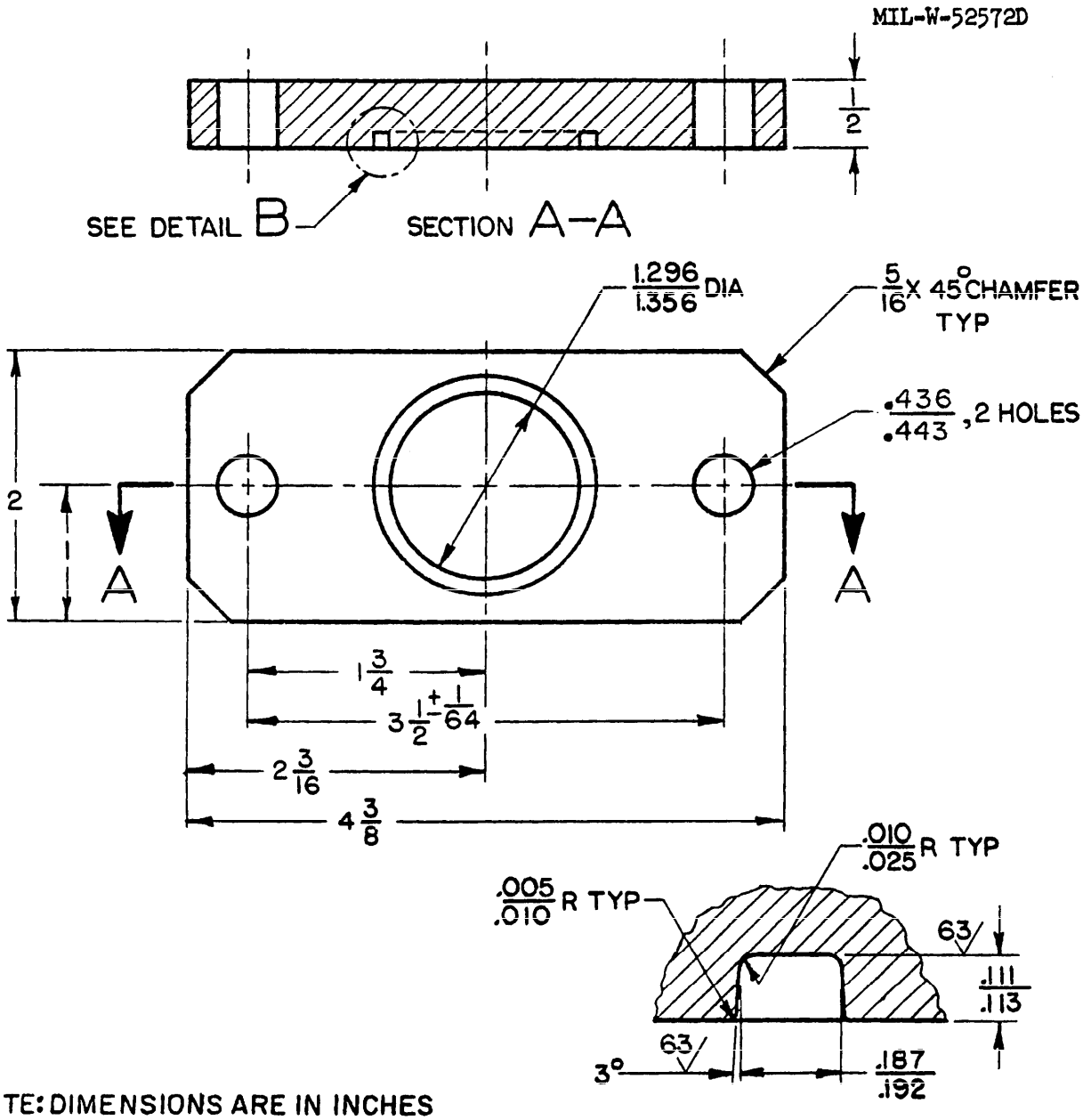
Review activity:
Army - MD

Preparing activity:
Army - ME

Project 4610-0087



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NOTE: DIMENSIONS ARE IN INCHES
UNLESS OTHERWISE SPECIFIED TOLERANCES
SHALL BE ± 1/32 INCH

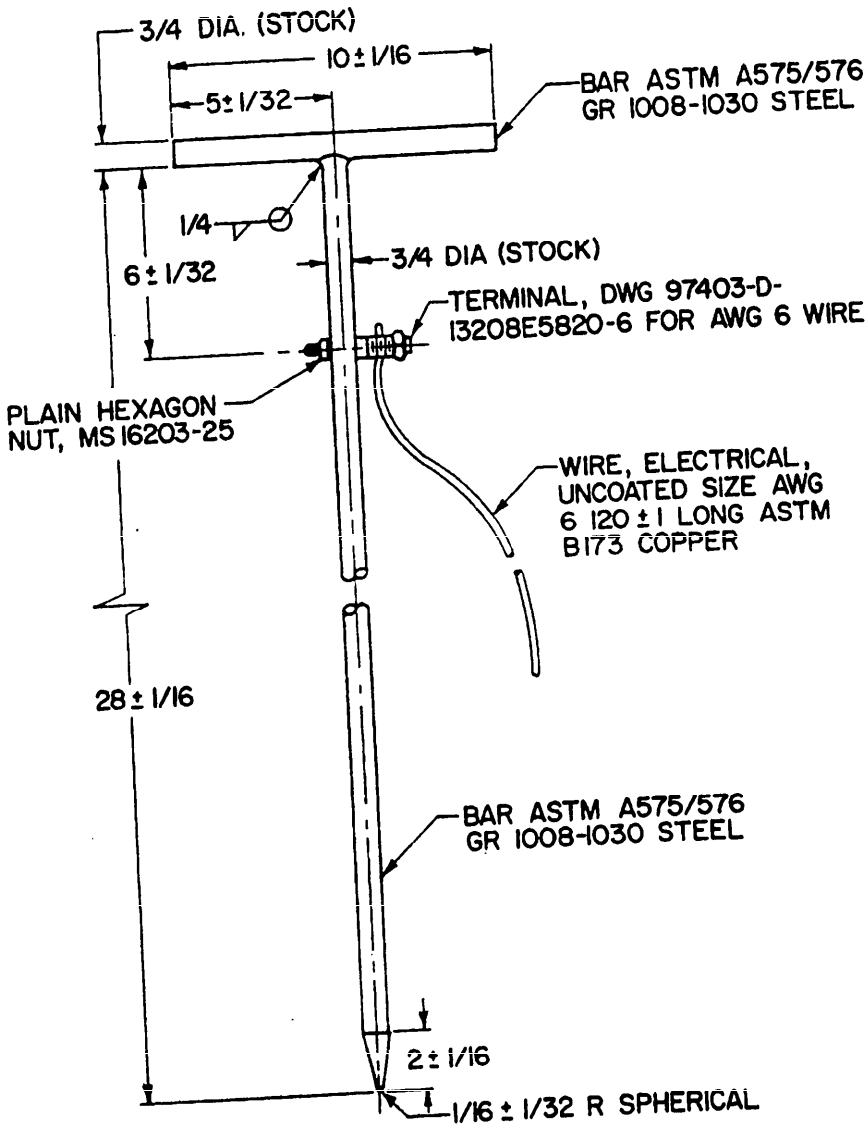
MATERIAL: ASTM-B209-6061-T6 ALUMINUM

FIG 1 FLANGE, BLIND



CX-1053

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NOTES:
1. DIMENSIONS ARE IN INCHES.
2. AFTER WELDING, HOT-DIP ZINC PLATE ALL PARTS, EXCEPT WIRE, TERMINAL, NUT, IN ACCORDANCE WITH ASTM-A123.

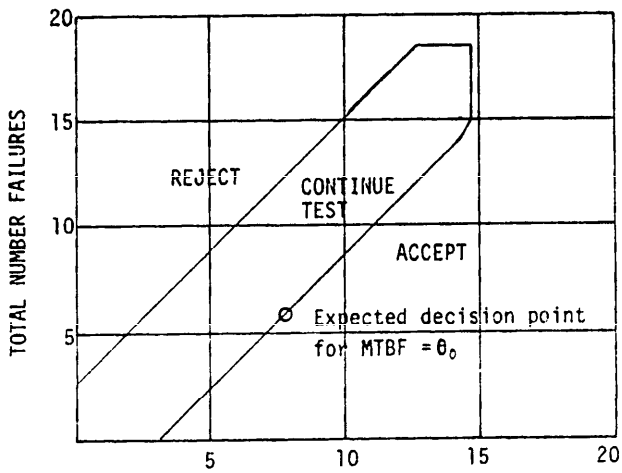
FIGURE 2. ELECTRICAL GROUND ROD.

DX290D



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Decision Risks 20%
Discrimination Ratio 1.5 : 1




| No. of Failures | Total Test Time* Reject (Equal or less) | Accept (Equal or more) | No. of Failures | Total Test Time* Reject (Equal or less) | Accept (Equal or more) |
|---|---|---|---|---|---|
| 0 | N/A | 2.79 | 10 | 5.84 | 10.90 |
| 1 | N/A | 3.60 | 11 | 6.65 | 11.71 |
| 2 | N/A | 4.41 | 12 | 7.46 | 12.52 |
| 3 | .16 | 5.22 | 13 | 8.27 | 13.33 |
| 4 | .97 | 6.03 | 14 | 9.08 | 14.14 |
| 5 | 1.78 | 6.84 | 15 | 9.89 | 14.60 |
| 6 | 2.60 | 7.66 | 16 | 10.70 | 14.60 |
| 7 | 3.41 | 8.47 | 17 | 11.52 | 14.60 |
| 8 | 4.22 | 9.28 | 18 | 12.33 | 14.60 |
| 9 | 5.03 | 10.09 | 19 | 14.60 | N/A |

*Total test time is total unit hours of "equipment on" time and is expressed in multiples of the specified MTBF. No single equipment "on" time shall be less than one half of the average operating time of all equipments on test.

FIGURE 3. Accept-reject criteria for reliability test.

X-3402



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-W-52572D Water Purification Unit, Electric-Motor-Driven, Erdlator-Type Clarifier, Diatomite-Type Filter, 420 Gallons Per Hour

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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DOD – 314



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PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research and
Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

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